# Asia Financials Credit Fundamentals

## An Introduction | October 2025

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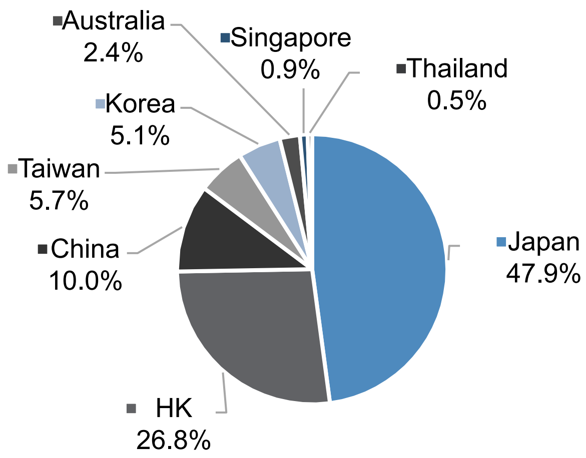
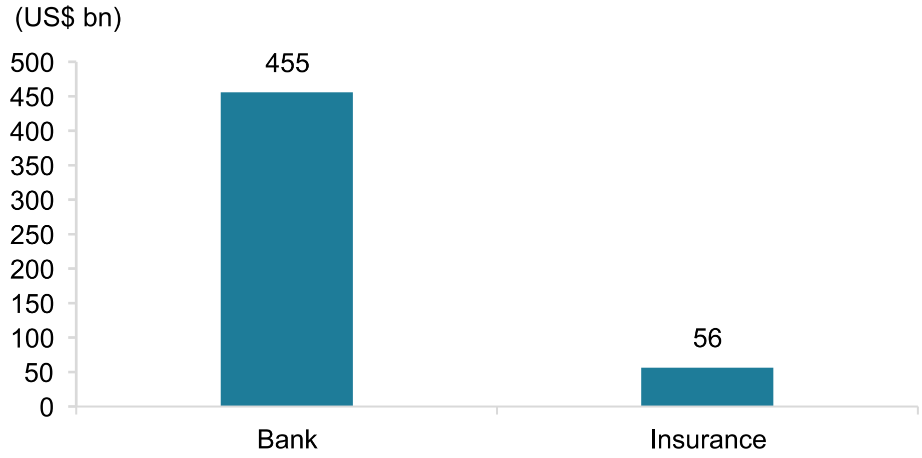
# Agenda

# Insurance – Asia credit market opportunity set

**MARKET CAPITALIZATION OF JACI APAC INSURANCE BONDS**




Source: J.P.Morgan.
Note: Market capitalization data as of Oct 16

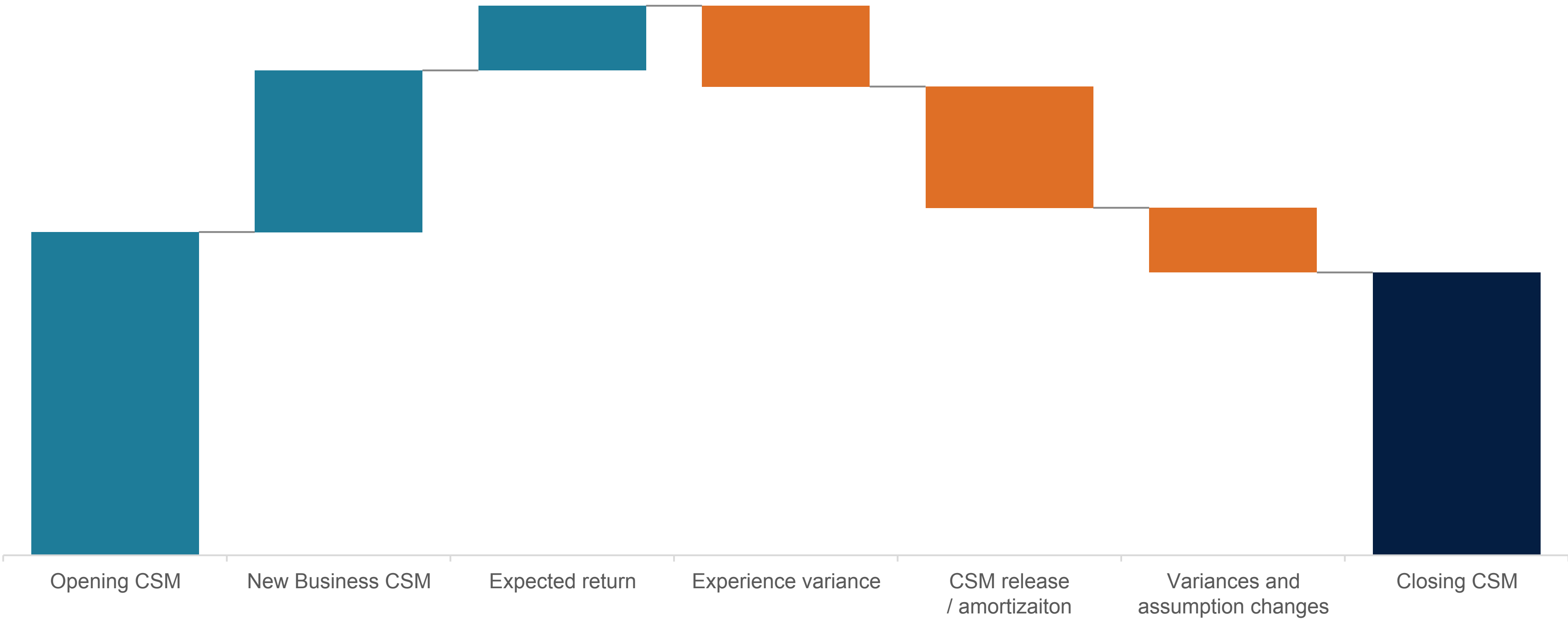
# Insurance – Simplified life insurance income statement

| | |
|---|---|
| **Insurance Revenue** | CSM release |
| **Incurred claims and expenses** | Expenses, including operating variances |
| **Insurance service result** | Net earnings from insurance activity |
| **Investment income** | Total investment income earned including policyholder |
| **Insurance finance expenses** | Unwind of discount rate and policyholder share of investment return |
| **Net investment result** | Consistent investment margin definition with non-life insurance |
| **P&L** | |

Source: J.P. Morgan.

# Insurance – Contractual Service Margin (CSM)

CSM – An estimate of future profit that will be earned over the life of an insurance contract




Source: J.P.Morgan.

# Insurance – Introduction to bond structures

Bonds may come in a variety of formats depending on the purpose and regulatory framework

- Why issue?
  - Senior instruments – funding
  - Subordinated instruments
    - Dated or perpetual
    - Can qualify as capital for regulatory solvency purposes
    - Ratings agency recognition in leverage calculations
- Generally, insurance bond structures across Asia are more diverse than bank capital instruments
  - Differences in regulatory frameworks across jurisdictions
  - Regulatory standards are evolving in some jurisdictions
- As with all credit analysis, we use the offering documents to understand details of a particular bond's terms & conditions

Source: J.P.Morgan.

# Insurance – Introduction to bond structures

## Example of subordination language

| | HLINSU 6.3% TIER 2 SUBORDINATED CAPITAL SECURITIES | TYANLI 6.25% TIER 2 SUBORDINATED SECURITIES |
|---|---|---|
| **Subordination** | • In the event of the Winding-Up of the Issuer, the rights and claims of the Securityholders to payment of principal and any Distribution under the Securities shall rank senior to, and in priority to, the claims of all holders of any present or future Junior Obligations of the Issuer, but shall be subordinated in right of payment to the claims of all other present and future senior and subordinated creditors of the Issuer (including, but not limited to, policyholders, general creditors and dated subordinated creditors of the Issuer), other than the claims of holders of Parity Obligations of the Issuer. The Securities will rank pari passu with Parity Obligations of the Issuer. | • The Securities constitute direct, unsecured and subordinated obligations of the Issuer and will at all times rank (i) junior to the Senior Obligations of the Issuer, (ii) pari passu and without any preference among themselves, pari passu with any Parity Obligations of the Issuer and (iii) senior to, in priority to the claims of holders of, Junior Obligations of the Issuer. The rights and claims of the Securityholders are subordinated as described in Condition 3.2. |
| **Junior Obligations** | • “Junior Obligations” means (i) any class of the Issuer’s share capital (including but not limited to any preference shares) and (ii) any other obligations of the Issuer that rank or are expressed to rank, by their terms or by operation of law, junior to the Securities. | • “Junior Obligations” means (i) Subordinated Capital Securities issued by the Issuer, (ii) any class of the Issuer’s share capital (including but not limited to any preference shares) and (iii) any other obligations of the Issuer that rank or are expressed to rank, by their terms or by operation of law, junior to the Securities.<br>• “Subordinated Capital Securities” means perpetual subordinated capital securities issued pursuant to the Regulation on Supervision of Insurance Business and its Detailed Enforcement Regulation, as they may be amended from time to time. |

Both bonds rank as “Subordinated” on Bloomberg’s DES screen, but TYANLI is senior to perpetual subordinated bonds.

Source: Offering documents.

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# Bank analysis – overview

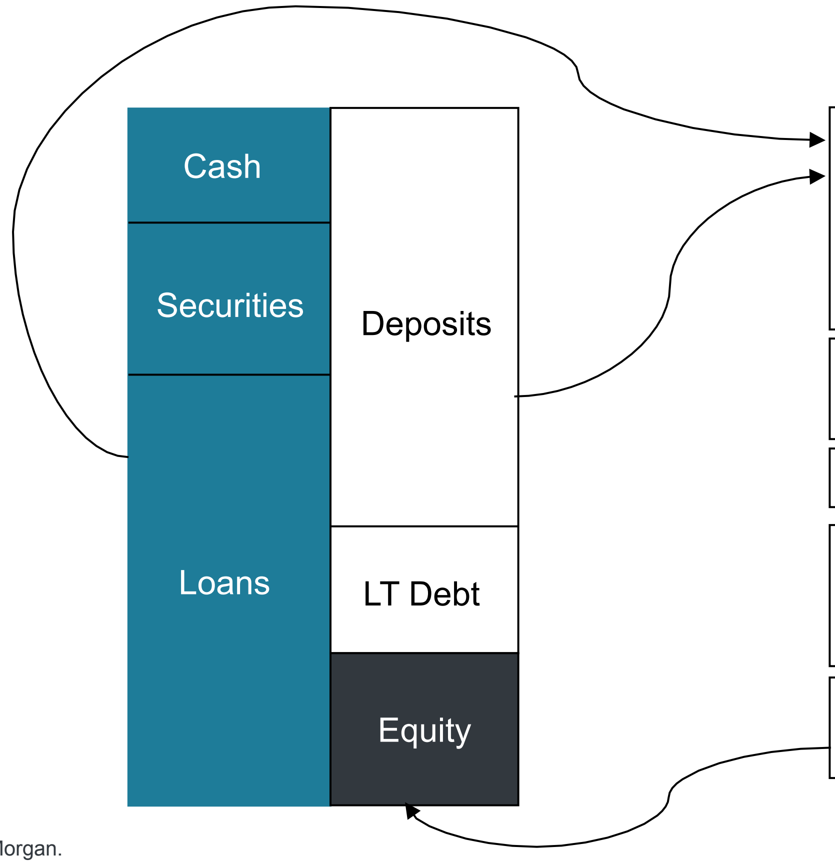
## What makes banks different?

| | BANKS | CORPORATES |
|---|---|---|
| **Regulation** | • Heavily regulated (Basel III, TLAC BRRD, etc.) to protect depositors, help market confidence and limit systemic risk | • Much less regulation, and only generally in terms of the prices and anti-trust |
| **Leverage** | • Business is highly levered by nature<br>• Regulators **limit the extent of leverage** through capital adequacy rules | • Varies between industries and players<br>• Controlled by the market and cost of capital rises with leverage |
| **Assets** | • Generally, highly liquid balance sheet, which can change in a short space of time.<br>• **'Too big to fail' banks** are perceived to receive implicit sovereign support even without any government ownership | • Balance sheets tend to be less liquid and more permanent in nature |
| **Analysis** | • Primarily focused on balance sheet (and income statement) | • Primarily focused on cash-flows (and income statement) |

Source: J.P. Morgan.

# Basic concepts – Balance sheet and income statement

**BALANCE SHEET**




**INCOME STATEMENT**

Interest income
Interest expense
**Net interest income**
Provisions against potential losses
Risk-adjusted interest income

Trading income
Fee income and other non-interest income

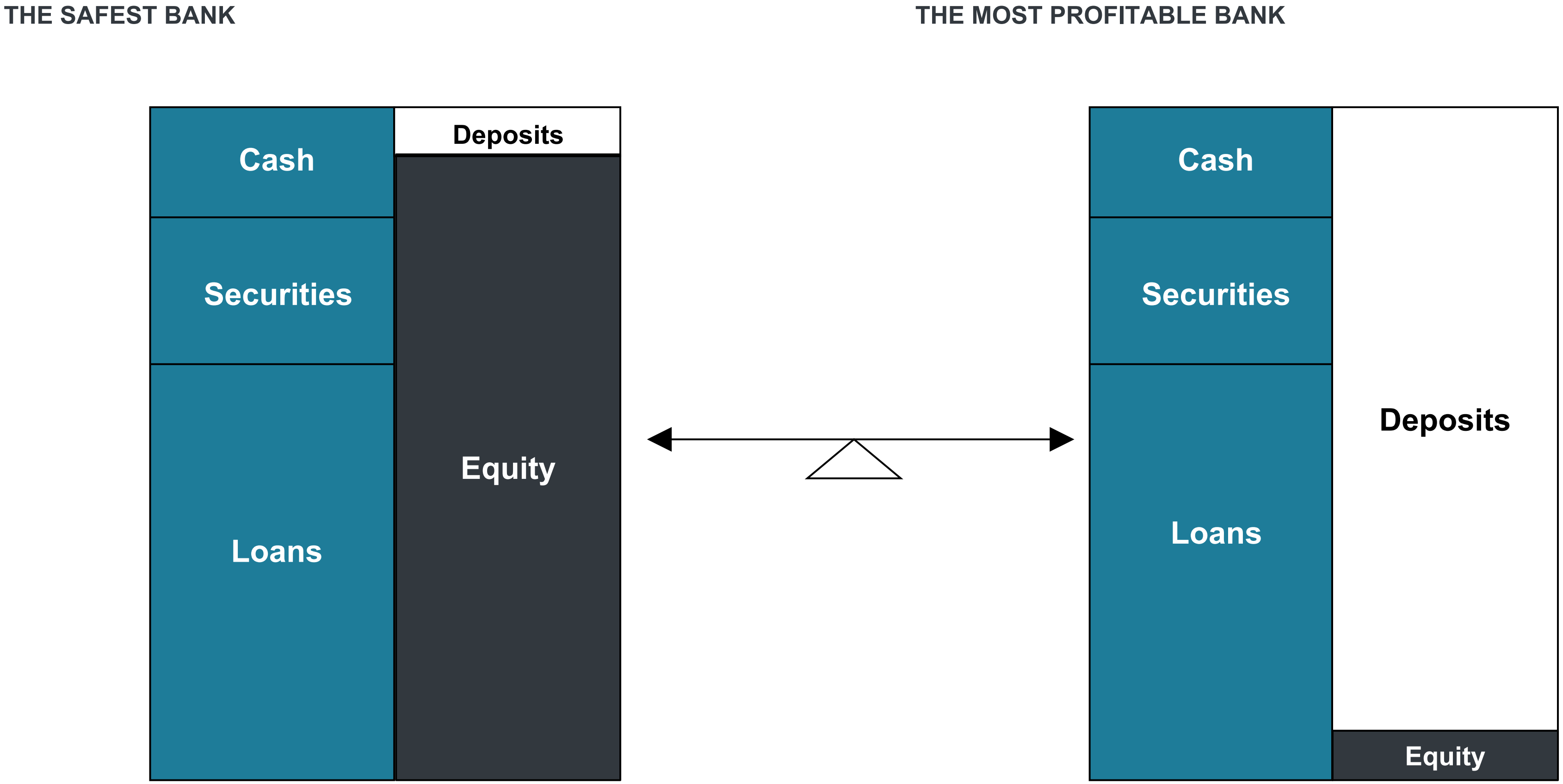
Operating expenses

**Operating income**
Tax
**Net income**

Dividends
Retained earnings

Source: J.P. Morgan.

# Basic concepts – Leverage




Source: J.P. Morgan.

# Basic concepts – The big picture

- What type of bank is this: retail, corporate, investment, universal (all three), bancassurance (bank + insurance)?
- What are its main business lines?
- Where is the bank domiciled and where else does it operate?
- Size of the bank, its domestic market share – is it important to the functioning of the banking system?
- Who are the shareholders?  Do they play an important role? Can they be relied upon to subscribe to more capital if needed?
- External credit ratings (S&P, Moody's, Fitch, local agencies)
- As a creditor, where do you stand in terms of structure or contract subordination? Is this an operating bank or a holding company? Is it senior or subordinated debt?

Source: J.P. Morgan.

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# Asset quality

## Principles

- A bank's assets mostly comprise interest-earning assets: inter-bank exposure, liquid debt securities, loans. Each interest-earning asset has some degree of risk of non-payment – i.e., credit risk
- Asset quality drives the risk-adjusted degree of profitability
- Analysis of asset quality focuses on loans but should also pay attention to the risks in the securities portfolio, as seen during the subprime crisis
- Qualitative factors to consider
  - Relative size of the loan book
  - Types of lending (secured vs unsecured, corporate vs retail)
  - Concentration risk: by geography, industry and individual borrower
  - Does the bank price risks correctly and assign appropriate reserves?
  - Quality of collateral and ability to claim the collateral if the borrower defaults, which also depends on the legal environment

Source: J.P. Morgan.

# Asset quality

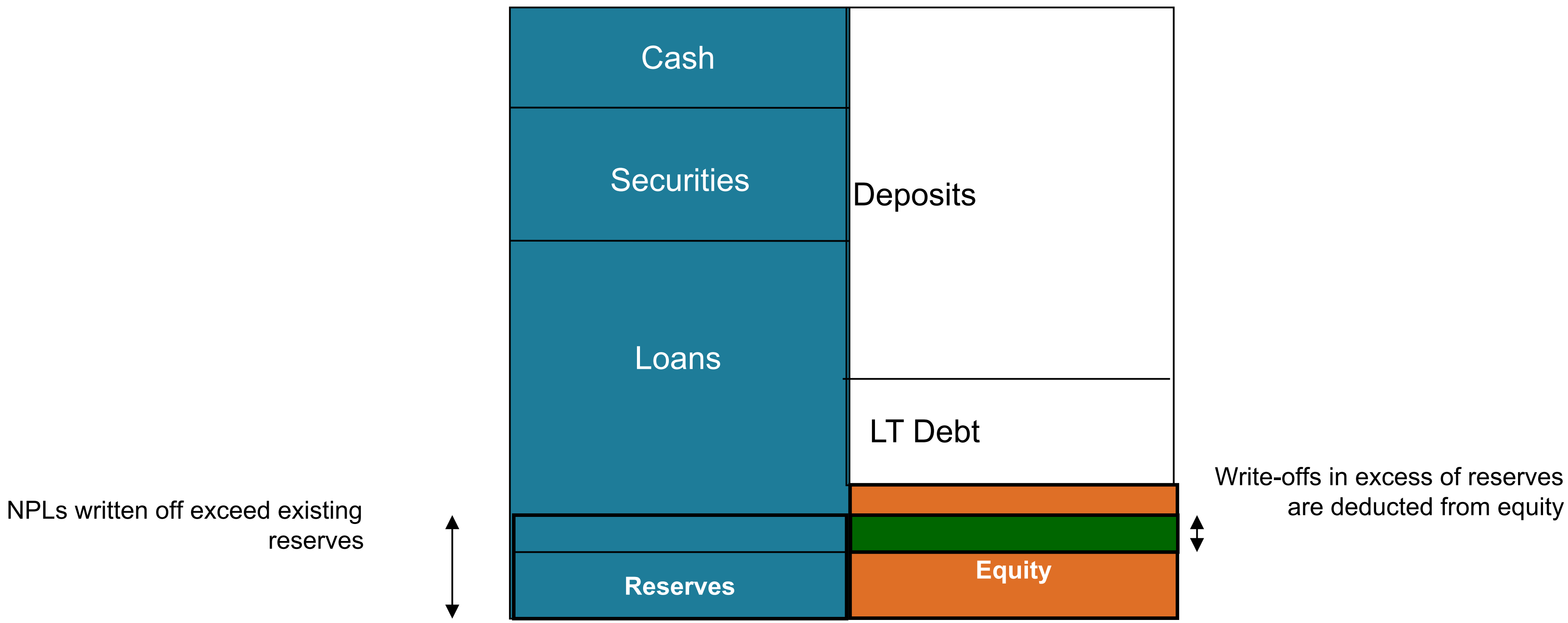
## Other considerations

- Every country defines non-performing loans differently
  - Also referred to as “bad loans” or “delinquent” or “past due”
  - Benchmark standard is 90+ days past due for interest or principle. Ideally is a forward-looking measure of borrowers’ ability to repay debt
  - Often classified into 3 sub-categories: substandard, doubtful, and loss
- Reserves are an important buffer against losses
  - Reserves against loan losses are a contra-account on the asset side
  - Banks make provisions (income) against losses annually, adding to the stock of reserves on the balance sheet
  - Bad loans are written off against their corresponding reserves
- Many regulators put a limitation on how long a “loss” asset can be left on the balance sheet e.g., if a loan is 12 months past due, it must be written off.
  - If an old NPL that was written off is eventually recovered, the bank recognizes it as a reversal of provisions and then reserves

Source: J.P. Morgan.

# Asset quality

**WRITING OFF BAD LOANS**




Source: J.P. Morgan.

# Asset quality

## Ratios

| | DESCRIPTION | COMMENT |
|---|---|---|
| **NPL ratio**<br>**NPL / Equity** | • Non-performing loans as % of gross loans<br>• NPLs net of reserves as % of equity | • If loans are growing rapidly, NPL ratio will look artificially low<br>• This is a measure of how much equity would be lost if the bank had to write off all of its bad loans once |
| **LLR / Gross Loans** | • Loan loss reserves (specific & general) as % of gross loans | • Over 100% is good but check whether net or gross of collateral |
| **Provision / PPOP**<br>**Provision / Avg. Loans** | • Provisions as % of pre-provision operating profit<br>• Provisions as % of average loans | • NPL ratio can also be understated if net charge-offs are very high<br>• Suggest bad loans are written off as fast as they are classified |
| **Net charge-offs / Avg. Loans** | • Net charge-offs (loans written off – loan recovered) % of average loans | • Can add back loans written off to show "true" NPL ratio<br>• If charge-off data is not available, look for very high rates of provisions |

Source: J.P. Morgan.

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# Earnings

- Does the bank make a profit? If not, where is the problem? Is it temporary or recent?
- Consider the revenue mix between interest income (from loans and securities), fees (from asset management, insurance, trade finance), and trading income (from securities portfolio). Are the components stable or volatile?
- What is the proportion of flexible to fixed expenses? Are the expense ratios in line with the size of the bank and the business areas in which it operates? Are expenses high relative to revenues?
- What is the level of loan loss provisions relative to operating profit? Does it look reasonable when you look at the character of the loan portfolio?
- Any exceptional/extraordinary items? Are there many one-time gains/losses? What is the tax rate and is it sustainable (e.g., is a low rate due to tax benefits)? What is the proportion of earnings paid out as dividends?
- In all of the above, think about the trend (past 3-5 years) and compare with peers

Source: J.P. Morgan.

# Earnings

## Ratios

| | DESCRIPTION | COMMENT |
|---|---|---|
| **Net interest margin (NIM)** | • Net interest margin = Net interest income / average earning assets. Core measure of profitability. In developed markets 1%-2%, in emerging markets 2.5%-5% is common. | • Core measure of profitability. In developed markets 1%-2%, in emerging markets 2.5%-5% is common.<br>• Can vary over the interest cycle if the bank is not able to adjust the rate it pays on deposits or borrowings, or the interest it earns from lending or buying securities |
| **Cost / Income** | • Cost/income ratio = Non-interest operating expenses / operating revenue. | • Below 45% is excellent, around 55% is average, above 70% is poor. To some extent explained by business model (capital versus labor intensive, distribution platform) and country |
| **Credit costs** | • Provisioning rate = Provisions / average loans. | • High levels often the cause of net losses. Can also be the result of regulators introducing stricter loan classification standards. |
| **ROA**<br>**ROE** | • Return on Average Assets (ROA) = Net income / Average total assets.<br>• Return on Average Equity (ROE) = Net income / Average shareholders' equity | • ROA: Over 1% is strong, under 0.5% is weak |

Source: J.P. Morgan.

# Agenda

# Liquidity

- Banks fund themselves with customer deposits, wholesale funding (bonds and interbank borrowing) and securitization.
- A stable retail deposit base is best, as savings and transaction (checking or current) accounts are “sticky” and people find it inconvenient to change their main banking account. In contrast, wholesale funding can rise in price quickly and can be very sensitive to the borrowing bank's reputation. Securitization limited by regulators, and being scaled back.
- Liquidity is the single most important factor for banks – ultimately banks fail because of lack of liquidity, not insolvency driven by weak asset quality.
- The regulator sets minimum standards for liquidity, which banks meet by holding low-yielding but liquid assets such as cash, government bills, and government bonds.
- Banks aim to minimize their mismatch between liquid asset and liabilities.
- Regulators often act as the lender of last resort in event of a liquidity crisis.

Source: J.P. Morgan.

# Liquidity

## Ratios

| | DESCRIPTION | COMMENT |
|---|---|---|
| **Deposit** | • Core deposits / Total deposits = current and savings deposits as % of total deposit base.<br>• Loans / Deposits | • A ratio of 40% or higher is good, indicates a stable consumer funding base<br>• A ratio over 100% indicates the balance sheet is over-leveraged. The risk is that the assets backing deposits are composed mostly of loans rather than less-risky government securities |
| **Interbank funding** | • Net interbank funding = interbank loans – interbank borrowing. | • Shows if a bank is borrowing from other banks to fund its balance sheet, which is generally less profitable. Banks may opt to deliberately run a gap to take advantage of rising or falling rates |
| **Liquid assets** | • Liquid assets/Total assets | • Will be mostly determined by regulators' requirements |
| **Foreign currency exposure** | • Net foreign currency exposure as % of equity<br>• Loans and deposits in foreign currency | • A measure of potential vulnerability to sudden changes in exchange rates, although if exposure is hedged may not be important<br>• if loans are not matched by foreign currency funding, ask what is the purpose of the FX loans |

Source: J.P. Morgan.

# Liquidity

## Regulatory measures

### LCR = LIQUIDITY COVERAGE RATIO

- A measure of short-term liquidity risk
- Requires banks to have a sufficient amount of unencumbered high-quality assets that can be easily converted to cash in a stress period

- The ratio of high-quality liquid assets (HQLA) to net outflows
- Minimum requirement is usually 100%

- HQLA examples include cash at the central bank, highly rated bonds

- Net outflows is an estimate of cash outflows, from deposits for example, over a 30-day period of stress

### NSFR = NET STABLE FUNDING RATIO

- NSFR is intended to promote stable funding over a longer time horizon
- Requires banks to fund balance sheets with stable funding sources

- The ratio of available stable funding (ASF) and required stable funding (RSF)
- Minimum requirement is usually 100%

- ASF, available stable funding increasing from equity, long-term debt, stable deposits, etc.

- RSF, required stable funding assesses the liquidity

Source: J.P. Morgan.

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# Other considerations: management

- Is there a coherent and realistic strategy? Is it expressed in terms of measurable goals? Does it generally achieve or consistently miss its targets?
- Does management have the necessary ability and experience? Has the bank expanded into new areas where it has little experience of assessing the risks?  Has it acquired businesses that it may not understand?
- Can management make decisions independently or is there pressure from related companies (corporate cross shareholding)?
- Does the bank operate free from regulatory pressure or government interference? If a bank does serve government goals, is there a mechanism to compensate the bank for any additional risks the bank faces as a result?
- Does the bank have a good name and a solid market franchise?
- Does the bank have a good relationship with its regulators? Has it been under investigation? Does it have a track record of bad surprises?

Source: J.P. Morgan.

# Other considerations: external factors

- **Regulatory environment**
  - Good regulation can put a floor under bank ratings by forcing banks to meet **high minimum standards**
  - Conversely, even the best run bank will be weakened by a poor regulatory environment. Other banks will be able to operate with weaker lending standards or higher leverage, making it difficult for a good bank to defend its standards
- **Legal framework**
  - Asset quality is highly influenced by the status of the legal system
  - Asset quality is enhanced if banks can easily claim any collateral related to defaulted loans
  - Countries with an efficient, predictable judicial system typically have higher recovery rates on collateral

Source: J.P. Morgan.

# Other considerations: bank credit ratings

- Key point: A bank that is strong in its own right does not always have a high rating
  - Close link between sovereign (government) rating and bank debt rating
  - Standalone rating = financial strength rating

|   |   |
|---|---|
|   | Standalone rating |
| × | Ability of government to support |
| × | Willingness to support |
| = | Debt rating |

- To gauge government's willingness to support a bank, look at share of deposits in the banking system, or importance to interbank loan market
  - Is the bank categorized as a domestic systemically important bank?
  - Would the failure of this bank cause distress to a significant proportion of the community?
  - Would the failure of this bank disrupt the payments system?
  - What is the bank's relationship to the government (ownership, policy role)?
- What is the government's ability to support a bank?
  - Is the fiscal position strong?
  - Can the bank regulators use non-monetary methods to help the bank? E.g., regulatory forbearance

Source: J.P. Morgan.

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# Overview

## Bank capital under Basel 3

- Basel 3 was introduced in response to the Great Financial Crisis to make banking systems safer. Like Basel 1 and 2, it is a collection of guidelines that are implemented nationally.
- The Basel Committee on Banking Supervision (BCBS) comprises representatives from 27 countries (mainly G20, and including India, Indonesia, and Singapore).
- Malaysia, Thailand and the Philippines are not represented in the BCBS but are part of the Basel Consultative Group, and adapt their domestic regulations accordingly.
- Because banks are regulated at the national level, local implementation may differ from what is recommended by the BCBS. Country-specific nuances are important to understand.
- What has Basel 3 changed?
  - Both higher capital requirements and stricter rules around the quality of capital. Basel 3 added new requirements for elements that must be deducted in the calculation of capital.
  - Introduced a capital conservation buffer, a countercyclical capital buffer and SIFI surcharges for systemically important banks.
  - New Basel 3-compliant capital instruments are replacing legacy issues, which are gradually being phased out.
  - Introduction of a Leverage Ratio which uses non-risk adjusted assets, as a blunt tool to restrict excessive leverage in the system.
  - Basel 3 has also added liquidity requirements: Liquidity Coverage Ratio (LCR ) and Net Stable Funding Ratio (NSFR)

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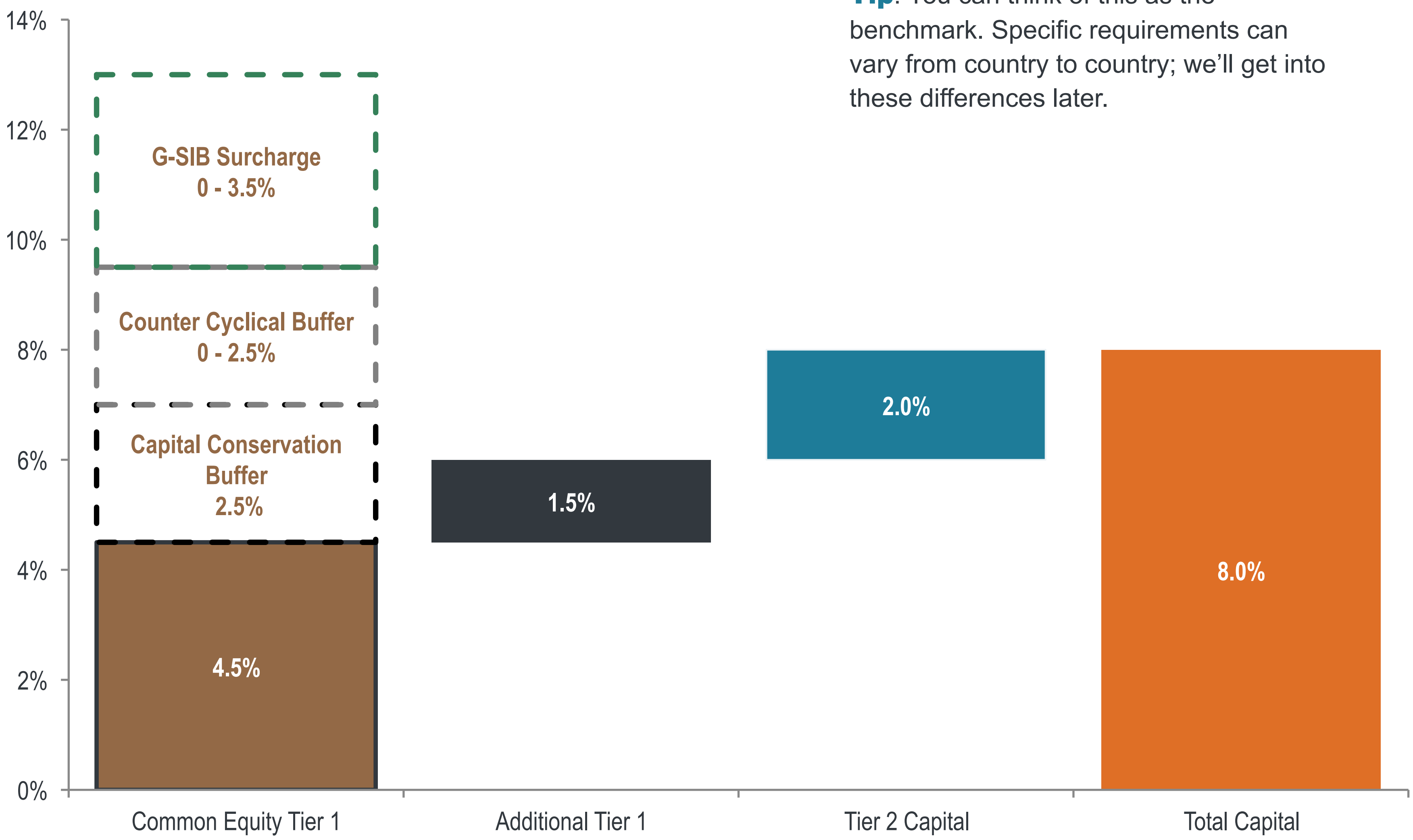
# The Basel 3 framework

## Basic capital requirements

**Tip**: You can think of this as the benchmark. Specific requirements can vary from country to country; we'll get into these differences later.

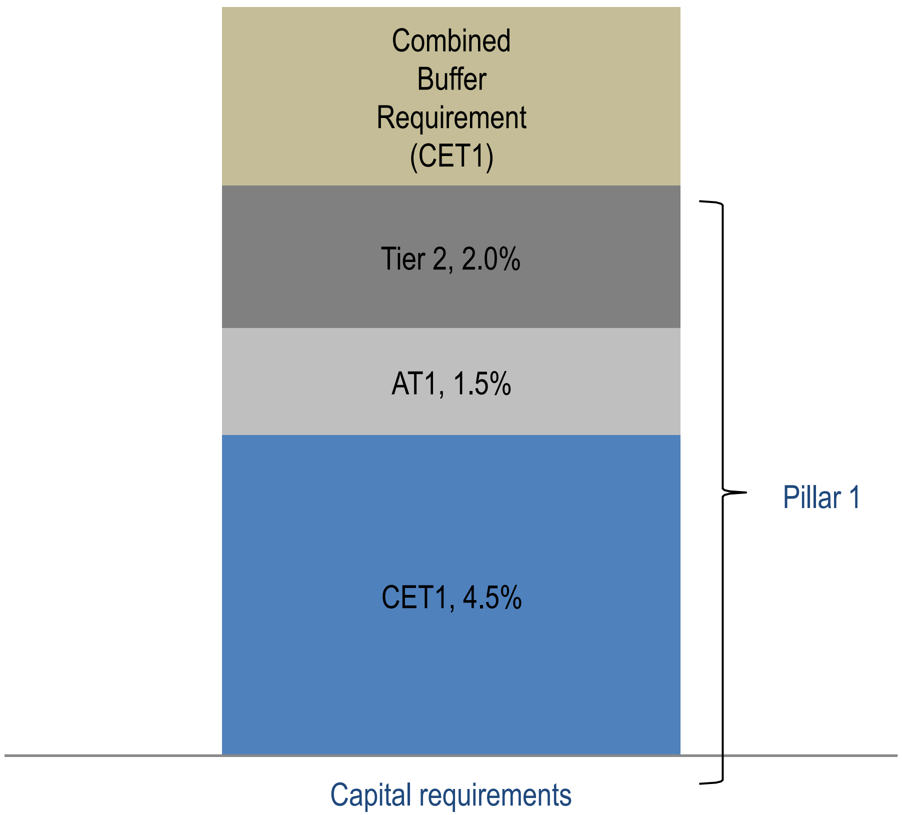



Source: J.P. Morgan.

# The Basel 3 framework

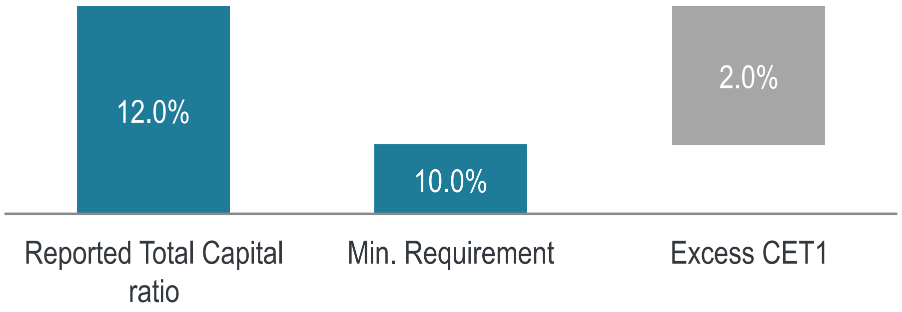
## Why issue AT1 / Tier 2?

**MINIMUM CAPITAL REQUIREMENTS**

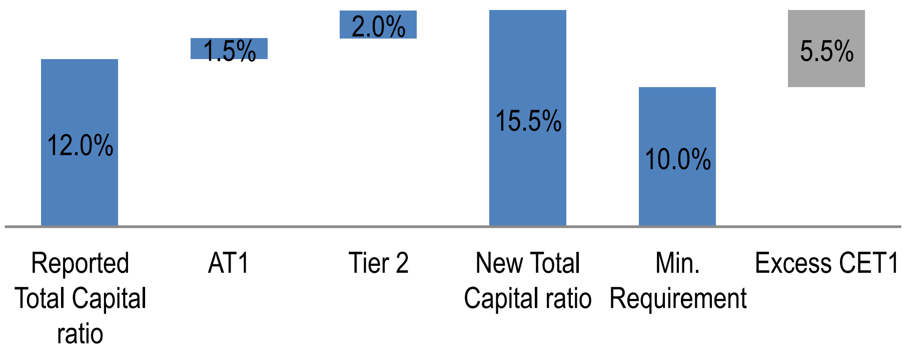



Source: J.P. Morgan and BCBS.

**WITHOUT AT1 / TIER 2**



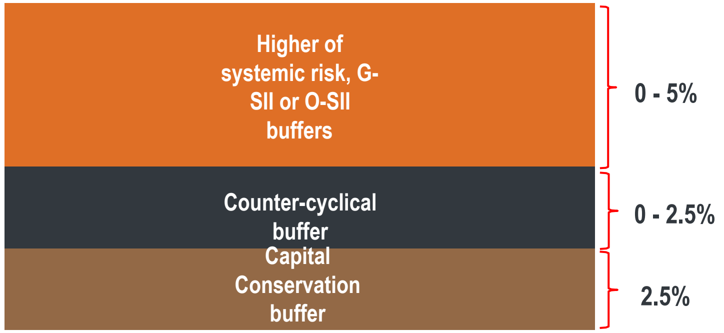

**WITH AT1 / TIER 2**



# The Basel 3 framework

## Combined Buffer Requirement (CBR)

**CBR COMPONENTS**




| | |
|---|---|
| **More stringent standards on capital adequacy** | In particular for SIIs, which go beyond the minimum Pillar 1 4.5% CET1 component and include additional capital buffers as a response to pro-cyclical effects of pre-crisis regulation |
| **High quality of capital** | The entire CBR must be met with capital of the highest quality; common equity Tier 1 (CET1) capital. |
| **Breach of CBR constrains discretionary distributions** | Not only ordinary dividends and incentive compensation but also **coupon payments for AT1 bondholders**. Excess of CET1 capital over the CBR is main indicator of deferral risk for AT1. |
| **Determinants of CBR** | Higher during periods of high credit growth and absorb losses in the downturn of the business cycle<br>Banks' individual characteristics, in particular the interconnectedness and systemic importance of a bank are one of the main determinants of the CBR |

Source: J.P. Morgan.

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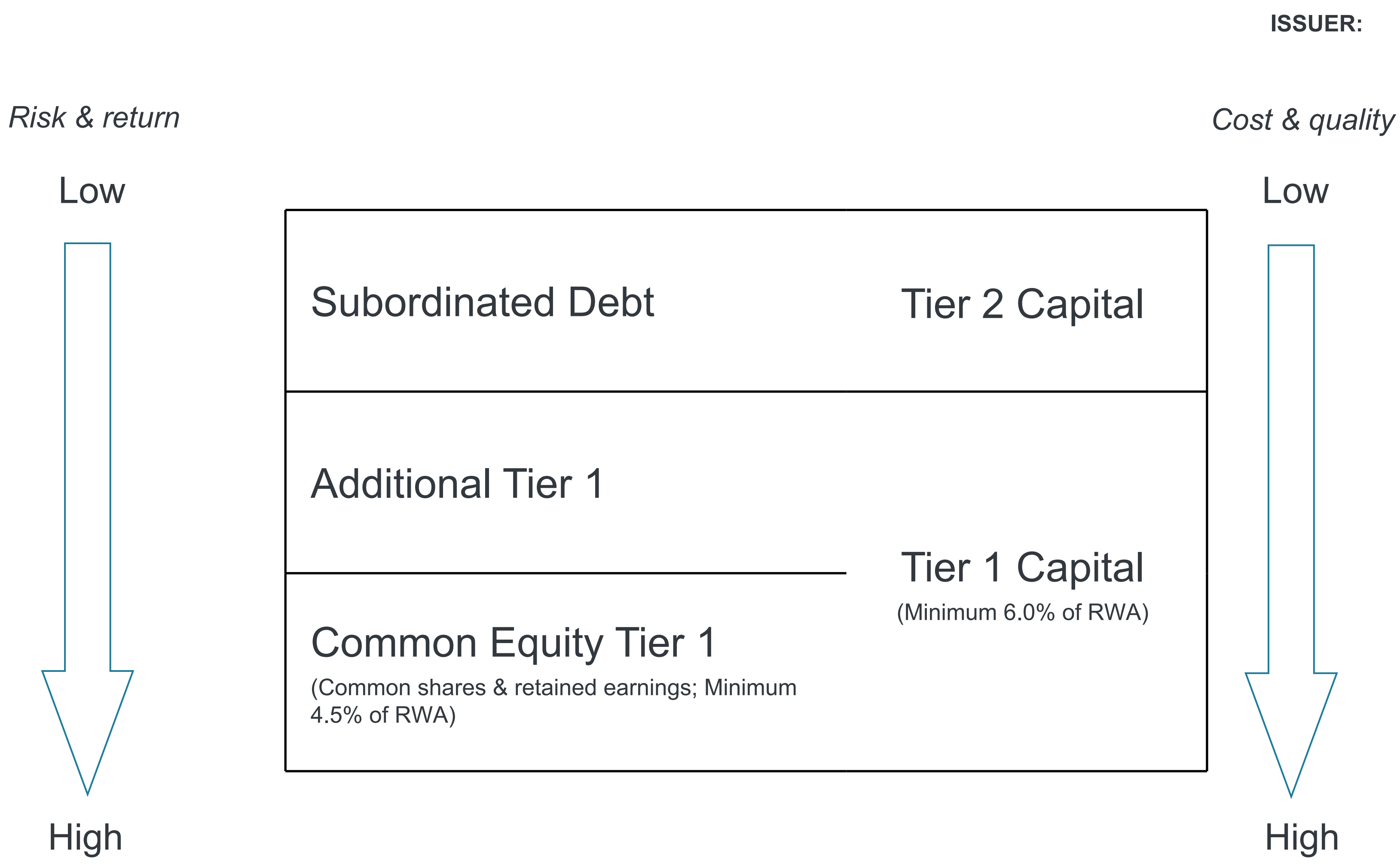
# Constituents of capital

## Structure is important

- During the GFC, many instruments that qualified as capital didn't actually function like capital.
  - Non-common equity elements of capital had loss-absorption that was limited to coupons. Principal could not be touched without wiping out equity.
  - Dividends / coupons that could be deferred were not.
- Under Basel 3, all elements of capital need to be loss-absorbing
- Loss absorption mechanisms
  - **Conversion to equity** – In Asia, these are mainly from China issuers.
  - **Write-down** – Most Basel 3-compliant instruments in Asia are in this format. China AT1 preference shares are an exception.
    - Permanent or temporary?
    - Partial or full write-down?
  - **Bail-in** is another consideration – is this statutory or contractual?
- Trigger for loss absorption at **PONV**.
- **Rank**. Is the Basel 3 instrument pari passu to legacy instruments, or actually subordinated?
- **Others**: coupon deferral, call language (standard, tax or regulatory?), dividend stoppers / pushers, frequency of call dates (1-time or multiple?)

# Constituents of capital

## Bank capital – the different tiers




Source: J.P. Morgan.

# Constituents of capital

## Capital considerations

| | Higher quality | | Lower quality |
|---|---|---|---|
| **Loss absorbency** | Able to absorb losses while the bank remains "going concern" | ⟺ | Able to absorb losses only under "gone concern" |
| **Availability** | Perpetual – no hard maturity | ⟺ | Dated – Hard maturity |
| **Costs** | Non-contractual Obligation<br>Deferrable, non-cumulative | ⟺ | Must Pay (i.e., failure to pay is an event of default) |
| **Subordination** | Deeply Subordinated | ⟺ | Less Subordinated |

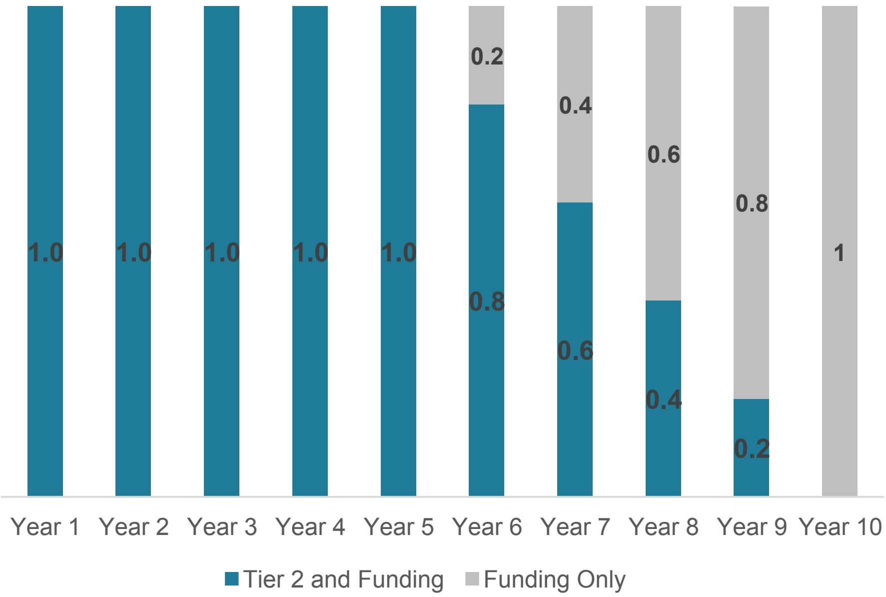
Source: J.P. Morgan.

# Constituents of capital

## Capital structure

### Tier 2 – The "lowest quality" capital

- Tenor: 5 years minimum
- Amount recognized as capital amortizes over last 5 years
- No incentive to redeem (such a coupon step-up).
- Subordinated to senior debt in event of default




Source: J.P. Morgan.

### Additional Tier 1 – More standardized definitions than under Basel 2

- Must have principal loss absorption: either conversion to equity or write-down.
- Perpetual
- May have a call after at least 5 years. Redemption is subject to regulatory approval.
- No incentive to redeem
- Dividends / coupons are fully discretionary, and cancellation cannot be an event of default.
- Senior only to common equity in event of default
- Loss absorption through either conversion to equity or write-down.

# Constituents of capital

## Point of Non-Viability (PONV)

- In order for a non-common equity instrument to qualify as capital under Basel 3, it must impose losses on investors at the point that at bank is deemed “non-viable.”
- The PONV is naturally an imprecise term. BCBS recommends the following triggers:
  - If capital ratios fall below some pre-specified level. BCBS has proposed CET1 < 5.125%.
  - Or, when the regulator deems the bank is no longer viable without using Basel 3-compliant instruments to absorb losses or without an injection of public sector funds.
  - So far in Asia, this is generally defined only at a regulator’s discretion (although China bank AT1s do have a hard capital ratio trigger of 5.125%).
- **Gone-concern** or **going-concern** capital?
  - instruments can be intended to be “going concern capital;” i.e., they would be triggered before resolution or liquidation.
  - Alternatively, they may only provide additional loss absorption only after a bank has effectively failed.

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# Additional Tier 1 (AT1)

## Main features

| | |
|---|---|
| **Loss Absorbency** | Write down: The bond can be fully/partially written down if the CET1 capital ratios falls below the trigger level<br>Conversion: The instrument is converted to equity at a fixed or floating price. |
| **Trigger level** | The trigger levels can reference either the consolidated entity, a subsidiary, or in some cases, both. There are bonds with low or high triggers:<br>Low: e.g., 5.125%<br>High: e.g., 7%, 7.125% or 8% |
| **Types of write-downs** | Temporary: The par value is written down temporarily and can be written back up<br>Permanent: No allowance for writing back up |

Source: J.P. Morgan.

# Additional Tier 1 (AT1)

## The call decision and extension risk

### Issuers have the option to call the AT1 at the first call date

- Basel 3 requires that there be **no incentive to redeem** the AT1 at any call date. This is to ensure the capital remains readily available if it is needed.
- AT1s must be replaced prior to redemption.
- Redemption requires approval from the regulator.

### Why an issuer would call:

- **Economics (a)**: Good "investor relations" may lead to favorable pricing for future issuance.
- **Economics (b)**: They can refinance (either onshore or offshore) with a less expensive instrument.
- **Signals** that the bank can access capital markets.
- Banking is a **confidence**-sensitive business; extension may raise fears of liquidity / solvency challenges if a call was expected.

### Why an issuer would not call (issuer extends):

- **Economics (a)**: Extension may raise the cost of future issuance if investors expect similar treatment in the future.
- **Economics (b)**: Refinancing would be more expensive than leaving the existing instrument outstanding.

# Additional Tier 1 (AT1)

## Minimum capital conservation requirements

**Maximum Distributable Amount**
of “profits” based on the common equity Tier 1 ratio

- 4.500 - 5.125% → 0%
- 5.125 - 5.750% → 20%
- 5.750 - 6.375% → 40%
- 6.375 - 7.000% → 60%

### Capital conservation requirements increase the risks of AT1

- Capital conservation requirements can restrict all distributions to providers of equity – including AT1 investors.
- Thus, both sufficient capitalization and earnings are necessary for AT1 distributions.
- Exception: Some banks in India had reported net losses and breached the capital conservation buffer, but they continued to make AT1 distributions.

# Additional Tier 1 (AT1)

What does the future hold?

## How well does it work?

- Instances of AT1 absorbing losses raise questions about how well it functions as intended
- Isolated write-down – a situation where common equity is not written down even as AT1 absorbs losses
- Ability to absorb losses on a “going-concern" basis?

## How to assess PONV risk?

- Instances of AT1 absorbing losses sometimes reflect a more general loss of confidence in the issuer by markets and / or regulators
- Some AT1 have suffered losses even when reported common equity capital were above minimum requirements plus buffers.
- Hard triggers may be too low to absorb losses on a “going-concern” basis

## Watch this space

- Governing bodies are reflecting on how well AT1 works
- Australian Prudential Regulation Authority (APRA) has proposed a phase-out of AT1
- A division within the Bank of International Settlements has published a thought piece that looks at the function of AT1 in recent examples, especially the problems associated with isolated write-downs.
- Remember: guidelines for the treatment of AT1, and for the regulatory approach to ailing financial institutions are always subject to implementation by local regulators

# Agenda

# Total loss absorbency capacity (TLAC)

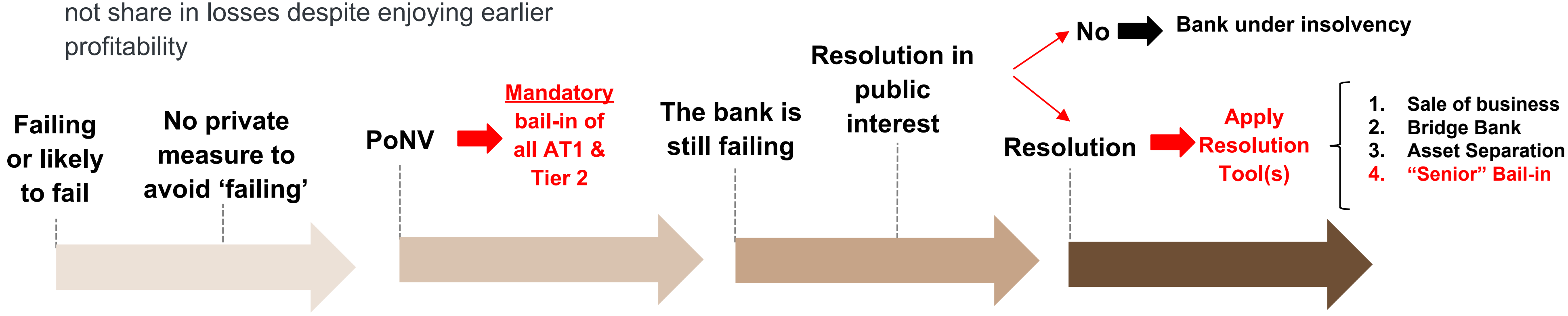
## The big picture – What happens when a large financial institution fails?

### The Challenge

- Traditional liquidations, such as through bankruptcy, may be problematic for systemically important financial institutions
- The failure of a large financial institution can pose the following problems that a corporate or small financial institution may not
  - Risks to financial stability
  - Risk of taxpayer losses if public support is required
  - Disruption of economic functions
  - “Moral hazard” – managers and creditors may not share in losses despite enjoying earlier profitability

### The Response

- The unique challenges posed by large financial institutions has led to a consensus about “tools” that should be available to regulators as part of an effective resolution regime
  - Sale of business
  - Bridge bank
  - Asset separation
  - Bail-in (TLAC)




Source: J. P. Morgan

# Total loss absorbency capacity (TLAC)

## The big picture – How?

**There are three broad ways of achieving TLAC eligibility**

**Structural**

Issued out of a holding company

**Statutory**

Legally subordinated to excluded liabilities

**Contractual**

Governing documents define subordination

**Features of TLAC bonds**

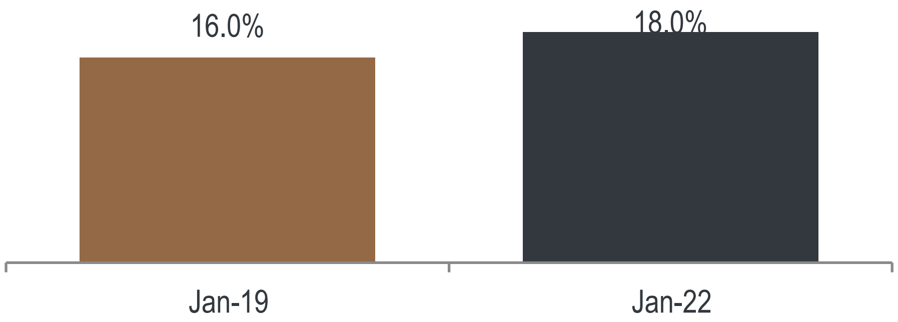
- No step-ups
- Dated
- No coupon or principal skip or deferral
- Not subject to write-down or loss-absorption outside or resolution or bankruptcy
- Excluded from TLAC when there is less than one year to maturity. This supports an optional issuer call one year before the final maturity

# Total loss absorbency capacity (TLAC)

## Background and requirements

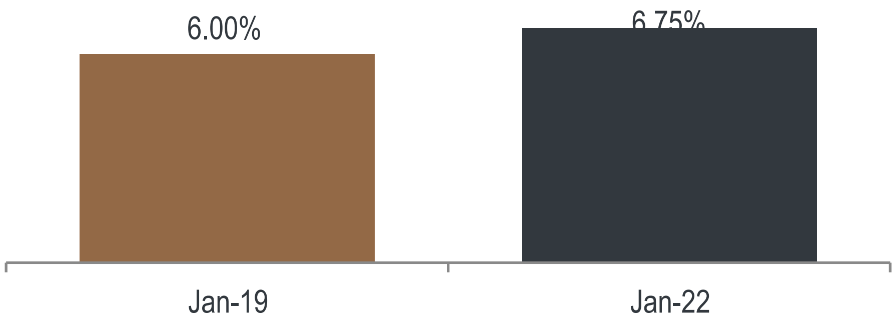
- The FSB ('Financial Stability Board'), the successor of the FSF ('Financial Stability Forum'), was established by the G20 in 2009 with the broad mandate to promote financial stability.
- In order to minimize state intervention to rescue systematically important banks and hence reduce the potential cost for the taxpayer, the FSB published in November 2014 the TLAC draft proposal. After a consultation period in the last few months the final term sheet was published at the beginning of November 2015.
- G-SIIs in DM countries have should meet a minimum TLAC requirement of at least 16% by January 2019 and increasing to 18% by 2022. Additionally, the leverage ratio will follow a similar implementation schedule; at least 6% by January 2019 and 6.75% in January 2022.
- CET1 used to meet the minimum TLAC must not be used to cover the CBR.
- The TLAC buffer can be cover with own funds (CET1, AT1 and/or Tier 2) and other eligible TLAC debt.
- TLAC requirements are particularly important for Tier 2 and senior unsecured issuance patterns.
- China is the only emerging market country with G-SIBs.

**TLAC RWA requirement (for DM)**




Source: J.P. Morgan and BCBS.

**TLAC leverage ratio requirement (for DM)**




Source: J.P. Morgan and BCBS.

# Agenda

# Regional implementation

**CHINA**

- Capital requirements are higher than Basel recommendations (minimum of 5.0% vs 4.5% under Basel standards.
- Capital conservation buffer: in line with Basel (2.5%). This was introduced ahead of the Basel timelines.
- Countercyclical buffer: no discussion yet.
- Leverage ratio: 4.0%, which is higher than the Basel's 3.0% recommendation.
- Resolution framework: Not yet implemented.
- TLAC – As an emerging market, China is exempt from TLAC requirements until end-2024.
- End 2024: 16% RWA / 6% leverage ratio minimum
- End 2028: 18% RWA / 6.75% leverage ratio minimum

Source: J.P. Morgan

**INDIA**

- Recurring examples of **forbearance**:
  - March 1, 2016: RBI amended details of the calculation of common equity Tier 1 capital, removing the need for immediate capital injections
  - February 2, 2017: RBI expanded the definition of distributable items to include statutory items
  - Regulator permits payments on local currency AT1 despite reported net losses and breaches of CCB.
  - 2018 – RBI directs banks under the Prompt Corrective Action (PCA) framework to redeem outstanding AT1s using "regulatory event" language.
    - Banks go under PCA because of poor capitalization, and need to conserve capital, not redeem or reduce capital.
    - The RBI may have wanted to avoid the risk of a coupon skip on an AT1.
- Examples of recurring **capital support** through regular capital injections from the Government or LIC

Source: J.P. Morgan.

# Regional implementation

**KOREA**

- PONV language provides more room for authorities for provide capital support than under standard Basel definitions.
  - PONV = Designation as an insolvent financial institution.
  - Regulators have legal space to provide capital and liquidity support to specific banks.
  - This is relatively investor friendly.
- Resolution planning – discussion have apparently stalled since the end of 2016; we await further updates.
- D-SIBs
  - 5 bank holding companies
  - 5 banks

Source: J.P. Morgan

**SINGAPORE**

- Capital requirements are stricter than Basel 3. Fully-phased in minimum ratios are:
  - Common equity Tier 1: 6.50%
  - Tier 1: 8.0%
  - Capital adequacy ratio: 10%
- Proposed resolution framework excludes senior creditors from bail-in.
  - This suggests a less aggressive approach to creditors in resolution.
  - Singapore and Hong Kong are the only Asia jurisdictions to be implementing a resolution framework that provides for bail-in.

Source: J.P. Morgan.

# Minimum capital ratio requirements

| | China | Hong Kong | India | Korea | Malaysia | Philippines | Singapore | Thailand | Japan |
|---|---|---|---|---|---|---|---|---|---|
| Capital Adequacy Ratio (CAR) * | 8.0% | 8.0% | 9.0% | 8.0% | 8.0% | 10.0% | 8.0% | 8.5% | 8.0% |
| Tier 1 Ratio (T1) * | 6.0% | 6.0% | 7.0% | 6.0% | 6.0% | 7.5% | 6.0% | 6.0% | 6.0% |
| **Buffers**** | | | | | | | | | |
| Systemically Important Banks(SIB) | 1.0% | 1.0 - 3.5% | 0.2 - 1.0% | 1.0% | N/A | 1.5 – 2.5% | 2.0% | 1.0% | 0.5 - 2.5% |
| Counter-cyclical Buffer (CCyB) | N/A | 2.5%** | 0.0% | 0.0% | N/A | N/A | 0.0%** | N/A | 0.01% |
| Capital Conservation Buffer (CCB) | 2.5% | 2.5% | 2.5% | 2.5% | 2.5% | 2.5% | 2.5% | 2.5% | 2.5% |
| Total buffer (SIB + CCyB + CCB) | 2.5 - 3.5% | 5.0 - 8.5% | 2.5 - 3.5% | 2.5 – 3.5% | 2.5% | 2.5 – 5.0% | 2.5 - 3.0% | 2.5 – 3.5% | 3.01 – 5.01% |
| **Minimum common equity Tier 1 (CET1) requirement*** | 5.0% | 4.5% | 5.5% | 4.5% | 4.5% | 6.0% | 4.5% | 4.5% | 4.5% |

Source: Local regulatory authorities, SNL Financial and J.P. Morgan.
Ratios are on a fully phased in basis.
* Excluding buffers.
** Refers to the domestic “benchmark” CCyB. However, banks are assigned individual ratios depending on CCyB requirements in jurisdictions to which they are exposed.

# China AT1

## Two paths lead to a full AT1 capital tranche

**CHINA BANKS CAN RAISE AT1 CAPITAL THROUGH EITHER PREFERENCE SHARES OR PERPETUAL BONDS**

| | Preference Share | Perpetual Bond |
|---|---|---|
| Ticker | BCHINA | BCHINA |
| Country | CH | CH |
| Amt Outstanding | US$6,500 | RMB40,000 |
| Issuer ratings | A1/A/A | A1/A/A |
| Ratings (M/S/F) | Ba1/BB/NA | NA/NA/NA |
| Issue dt | 10/23/2014 | 1/29/2019 |
| nxt Call dt | 10/23/2019 | 1/29/2024 |
| Other Call dts | Every dividend payment date after | Every dividend payment date after |
| Maturity | Perpetual | Perpetual |
| Cpn | 6.750 | 4.500 |
| Reset | 5yT + 546bp | 5y CGB + 157bp |
| Host | Preference share | Capital bonds |
| Structure | Perpnc5 | Perpnc5 |
| Crncy | USD | CNY |
| Rank | AT1 | AT1 |
| Conversion | Conversion to H shares (@ HK$3.44) | Partial write-down |
| Write-up? | N/A | N/A |
| Trigger | CET1 =< 5.125%<br>PONV (regulators' discretion) | CET1 =< 5.125%<br>PONV (regulators' discretion) |
| Coupon deferral? | Non-cumulative<br>Issuer or regulator discretion | Non-cumulative<br>Issuer discretion |
| Mandatory deferral? | Yes, 1) must have distributable after-tax profits after offsetting the previous years' losses and contributing to statutory reserve fund and general reserves, AND 2) Meeting regulatory capital requirements. | N/A |
| Dividend stopper? | Yes, until a coupon has been paid on the AT1 securities. | Yes, until a coupon has been paid on the AT1 securities. |
| Early redemption | No | No |

Source: Offering document, J.P. Morgan and Bloomberg Finance L.P. Note: Ratings key: M/S/F = Moody's, S&P and FitchRatings.

# Agenda

# AT1 vs USD Preferreds

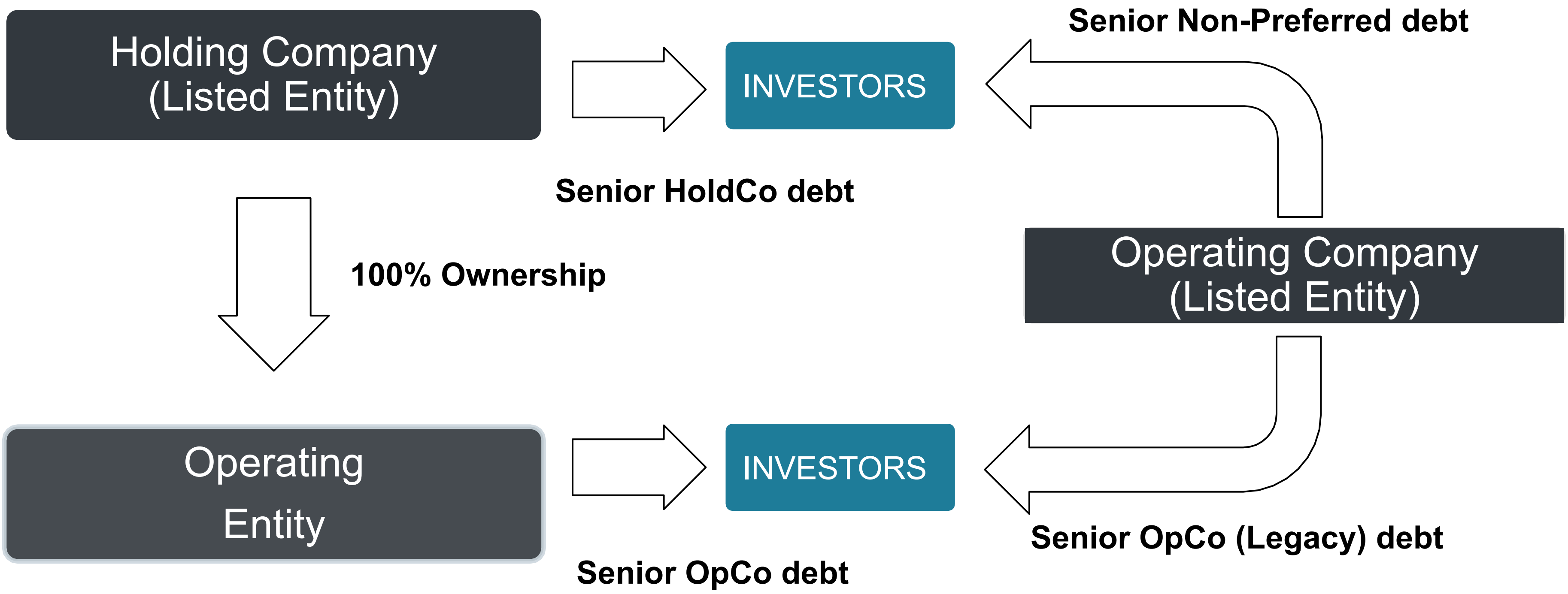
## Different forms of non-common equity Tier 1 capital

| Asia AT1 | European AT1 | US Preferreds |
|---|---|---|
| • Dividend stoppers and pushers are permitted<br>• $100 par and only issued for institutional investors (i.e. no retail deals)<br>• Fixed-to-Reset rate structure. Reset spread = issue spread due to ‘no incentive to redeem’ when the AT1 is first issued<br>• NC5 (so far; have not yet seen other structures)<br>• May have an explicit trigger for write-down or only a “soft” trigger<br>• Call / extension risk motivated by reputational, signaling considerations ahead of economic considerations. | • Neither dividend pushers nor stoppers permitted<br>• $100 par and only issued for institutional investors (i.e. no retail deals)<br>• Fixed-to-Reset rate structure. Reset spread = issue spread due to ‘no incentive to redeem’ when the AT1 is first issued<br>• NC5+ - more recently have seen more prevalence of NC10<br>• Quantitative trigger for write-down or equity-conversion | • Dividend stoppers permitted<br>• $25 par also issued – predominantly for retail investors. These are mostly fixed for life<br>• Fixed-to-Floating (Libor) or Fixed-to-Fixed Reset (Swaps) rate structure<br>• NC5 or NC10 structure, latter mostly for institutional investors<br>• No explicit quantitative trigger for write-down or equity-conversion<br>• Call/extension risk driven by economics (eg. prevailing rates) |

Source: J.P. Morgan.

# Topics for Europe and the UK

## Senior Unsecured Debt: Different Corporate Structure




Source: J.P. Morgan European credit research.

# Topics for Europe and the UK

## The TLAC problem

External TLAC must be subordinated to "Excluded Liabilities

"No Creditor Worse Off"

**ELIGIBLE EXTERNAL TLAC MUST NOT INCLUDE:**

a. insured deposits;

b. any liability that is callable on demand without supervisory approval;

c. liabilities that are funded directly by the issuer or a related party of the issuer, except where the relevant home and host authorities in the CMG agree that it is consistent with the resolution strategy to count eligible liabilities issued to a parent of a resolution entity towards external TLAC;

d. liabilities arising from derivatives or debt instruments with derivative-linked features, such as structured notes;

e. liabilities arising other than through a contract, such as tax liabilities

f. liabilities which are preferred to normal senior unsecured creditors under the relevant insolvency law; or

g. any other liabilities that, under the laws governing the issuing entity, cannot be effectively written down or converted into equity by the relevant resolution authority.

| | |
|---|---|
| **Statutory** → | **Germany** |
| **Contractual** → | **French banks (senior non-preferred debt)** |
| **Structural** → | **UK/Swiss Banks (senior HoldCo debt)** |

Source: J.P. Morgan European credit research and BCBS.

# Risk-weighted assets (RWAs)

| | € bn | RW | RWA € bn |
|---|---|---|---|
| Interbank Deposits | 25 | 20% | 5 |
| Trading Assets | | | |
| Sovereign Debt | | | |
| Italy (Baa2/BBB) | 10 | 0% | 0 |
| France (Aaa/AA) | 10 | 0% | 0 |
| Corporate Bonds | | | |
| Nestle (Aa2/AA-) | 20 | 100% | 20 |
| Ford (Baa2/BBB) | 20 | 100% | 20 |
| Loans to Customers | | | |
| Mortgages | 80 | 50% | 40 |
| Unsecured Lending | 25 | 100% | 25 |
| Total | 190 | | 110 |

Source: J.P. Morgan European credit research and BCBS.

- RWA calculations include capital requirements for credit risk, market risk and operational risk.
- **Standardized approach**
  - Simplistic methodology that assigns risk weights based on entity classification (corporate, sovereign, etc.)
  - Has several shortfalls, including:
    - Overreliance on external credit ratings
    - Lack of granularity
- **Advanced models**
  - Addresses some perceived weaknesses by using a more granular approach in terms of determining risk weightings
  - Based on actual PD ('probability of default') and LGD ('loss given default'). Internal Ratings Approach

# Agenda

Page

# Acronyms and definitions (1)

- **BCBS**: Basel Committee on Banking Supervision – The group of regulators who have designed the Basel 3 capital and liquidity frameworks.
- **CBR**: Combined buffer requirement – As a response to pro-cyclical effects of pre-crisis regulation, the BCBS ('Banking Committee on Banking Supervision') adopted more stringent standards on capital adequacy, in particular for systematically important institutions, which go beyond the minimum 4.5% CET1 component and include additional capital buffers. Includes the CCB, CCyB and SRB.
- **CCB**: Capital conservation buffer – The CCB is fixed at 2.5% of RWA for all institutions and, as for all other capital buffers; it should be covered entirely and only with CET1 Capital and increases the minimum CET1 capital ratio for all banks to 7%
- **CCyB**: Countercyclical buffer – The CCyB ensures that a bank has sufficient capital, accumulated during periods of high lending growth, to absorb losses during periods of high stress in the financial sector.
- **D-SII(B)**: Domestic systemic important institution (bank)
- **EBA**: European Banking Authority
- **ESRB**: European Systemic Risk Board
- **FPC**: Financial Policy Committee
- **FSB**: Financial Stability Board

# Acronyms and Definitions (2)

- **G-SIB**: Global systemic important institution
- **MREL**: Minimum requirement for own funds and eligible liabilities
- **MDA**: Maximum distributable amount – Restrictions imposed on distributions based on the quartile within the CBR in which the CET1 ratio of the bank falls
- **NCWO**: No creditor worse off – Clause in the Bank Resolution and Recovery Directive (BRRD) which states that as a result of resolution, no creditor should be worse off than they would have been in insolvency.
- **O-SII**: Other systemic important institution
- **SPE**: Single point of entry – A resolution regime in which all losses are transferred to and borne by the HoldCo first
- **SRB**: Systemic risk buffer – Systematically important banks, both G-SIBs and D-SIBs, have to maintain an additional CET1 capital buffer as a failure could jeopardize the financial system stability and result in a disproportionally larger impact on the wider economy
- **SRP:** Supervisory review process. Conducted by the Hong Kong Monetary Authority as part of the Pillar 2 requirements.
- **TBTF**: Too big to fail
- **TLAC**: Total loss absorbency capacity – Minimum requirement to ensure that G-SIBs have the loss-absorbing and recapitalization capacity in resolution and do not require public funds

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